Surveillance Over Time:
Automated Licence-Plate Readers on American Commutes, and the Rising Price of Avoiding Them

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Abstract

Automated licence-plate readers (ALPR) have spread from tollbooths to residential streets. We measure what that means for an ordinary trip, and how fast it is changing. A continental routing engine models each camera's field of view and can find a route that crosses none. We route 56,131 home–work commutes from all fifty states and the District of Columbia, with and without avoidance, past 186,316 installations: 142,257 mapped by volunteers in OpenStreetMap, and 44,059 more in service that only the registry of the dominant vendor, Flock Safety, records. 78.6 % of commuters pass at least one reader on the way to work. Refusal is still cheap: 80.8 % can reach work past none, and avoiding adds a median of 2.29 min—about two-fifths more than the volunteer map alone implies. OpenStreetMap keeps every edit, so we also rebuild its camera map as it stood on the first of every month since January 2024, node for node, and re-ask the question of each on the same commutes and roads. The map grew 128-fold in thirty-three months, and exposure almost in proportion: the cameras an average commute passes rose 94-fold, and the share of commuters passing one from 2.2 % to 75.3 %. Refusal grew dearer per camera as well as in total: on the first map of 2024 95 % of exposed commuters could reach zero, for a median of 0.8 min; on today's map 79 % can, for 3.1 min, and 75 % once the vendor's registry is added, for 3.9 min. What a commute passes, and what declining costs, are functions of a map that is still filling in; we say when ours was drawn.

Keywords: automated licence-plate recognition; surveillance; route choice; volunteered geographic information; OpenStreetMap; edit history.

1 Introduction

Automated licence-plate readers (ALPR) photograph every passing vehicle, resolve the plate, and retain the read [14]. Unlike a traffic camera they are not triggered by an offence, and unlike a patrol officer they do not tire. Fixed, pole-mounted units have spread from limited-access highways into residential neighbourhoods, arterial roads, and the entrances of private subdivisions [27, 40].

The public debate about this expansion is conducted in the vocabulary of policy: retention windows, warrant requirements, audit logs, data-sharing agreements. What that vocabulary lacks is a measurement of the thing being regulated. How much of this does a person actually drive through on the way to work on an ordinary Tuesday? And if they would rather not, is declining a practical option or a theoretical one?

This paper answers both, and does so by building the instrument the question implies. We construct a routing engine that treats ALPR fields of view as first-class geometry and can be asked

for a route that avoids them. Running each sampled commute through it twice—once normally, once avoiding—yields the exposure and the price of refusing it, for the same trip, on the same network, at the same speeds.

The second quantity is where the paper earns its keep. A privacy protection that costs four minutes is a feature. One that costs forty is available only to people with forty minutes to spare. We therefore report the price of refusal as a distribution rather than a point: for most commuters it is a few minutes, and for a tail it is not.

The third question follows from the data source, and it is the one this paper is about. The cameras are mapped by volunteers, and the map has grown 128-fold since January 2024. OpenStreetMap keeps every edit, so the measurement can be repeated on the map as it stood at any past instant. We repeat it on the first of every month for thirty-three months: the same commutes on the same roads, re-scored against each month's cameras and re-routed against each quarter's. How much surveillance a commuter drives through, and what declining it costs, turn out to be functions of how much of it had been mapped—and both have risen since, exposure on all but one of the monthly maps and the cost on every routed map from July 2024.

A volunteer map is also a lower bound, and for the present it can be checked against the vendor's own records. Flock Safety's device registry as it stood in December 2025 has been published; merged with the map, it adds 44,059 in-service readers in the fifty states and the District that no mapper had reached, nearly a third again. The current figures in this paper rest on that merged map. The trend, which a registry without dates cannot join, rests on the map alone, and the difference between the two is itself a measure of what the volunteer map still misses.

1.1 Contributions

- **A directional exposure model at continental scale.** Cameras are wedges, not circles. 98.0 % of mapped North American ALPR installations carry a bearing tag, making this feasible nationally; proximity counting would charge drivers for cameras aimed at the opposite carriageway.
- **Avoidance routing with no forked router.** Camera fields of view—the map's 157,084 cones, merged where they overlap into 135,210 polygons, and with the registry's 44,176 discs 159,286—are compiled into a stock GraphHopper spatial index at import time, so an ordinary per-request custom model can reference them by name.
- **A commute-weighted sample.** 56,131 real home–work pairs in all fifty states and the District of Columbia, drawn from LEHD LODES with probability proportional to worker count and weighted to each state's commuters.
- **The camera map at any instant, from the edit history.** Every version of every node ever tagged as a reader, extracted once from OpenStreetMap's full-history dump and validated node for node against the public history server, gives the map on the first of every month since January 2024 and the day each camera entered it. The map's coverage is checked against the counts 1,050 police departments publish, of which it holds a median 98 %.
- **The map, merged with the vendor's own registry.** Flock Safety's device records as they stood in December 2025, paired one-to-one with the map within 50 m: 88,754 registry devices stand on a mapped node, and the 44,059 in-service plate readers in the fifty states and DC that the map lacks enter the current picture as 60 m discs, the registry recording no bearing. They raise the price of refusal by two-fifths. The current figures rest on this merged map; the trend, which an undated registry cannot join, on the map alone.
- **A thirty-three-month trend on a fixed sample.** The same commutes on the same roads

against thirty-three monthly maps, re-scored for exposure on each and re-routed for avoidance on each quarter's: exposure rose almost in proportion to the map, and refusal grew dearer per camera as well as in total.

2 Related work

This paper sits where three bodies of work meet: research on ALPR as a policing technology, on volunteered geographic data and its biases, and a small practice of routing around surveillance. We take them in turn, and then state what is, to our knowledge, new here.

ALPR as a policing technology. Criminological research on ALPR has mostly asked whether it works: Lum et al. [26] tested ALPR-equipped patrols in a place-based randomized design, Lum et al. [28] review the impact studies since, and Shjarback and Sarkos [39] evaluate Atlantic City's fixed readers on every entrance and exit to the city. Sociological and legal work asks what the technology does to policing itself—plate readers lowering the threshold for inclusion in police data [2], the automated suspicion they enable [12, 24], and Flock's network as an infrastructure through which policing crosses jurisdictions [31]—and the legal status of the location records they produce is unsettled [3, 14]. None of this work measures how much ALPR surveillance an ordinary trip passes, or how fast that is changing.

Volunteered geographic data. The camera map is volunteered geographic information [19], whose quality has been assessed against authoritative data since OpenStreetMap's early years [21] and whose completeness is known to vary with place and development [22]. For surveillance specifically, the Atlas of Surveillance records which agencies use which technologies but not where devices stand [9]; DeFlock supplies the locations [4]. Coverage bias is the central threat to every result built on such data, and we know of no earlier measurement of it for ALPR; we compare the map with the camera counts police departments publish in Flock's transparency portals (Section 8.2), and its own history shows how far it was from complete on any earlier date.

Exposure along the way. That exposure to a place-bound feature depends on where people go and not only on where they live is the uncertain geographic context problem [25]; measuring surveillance along the commute rather than at the home address is an answer to it, and the unit of analysis throughout is the trip.

Routing around surveillance. A route that avoids cameras is an older idea than ALPR: in 2001 iSee computed "paths of least surveillance" around Manhattan's closed-circuit cameras [23, 38], of which a civil-liberties survey had counted some 2,400 visible from the street [32]. Several services now route drivers around mapped ALPR, some using each camera's field of view [1, 5, 6, 15], and in network measurement Edmundson et al. [7] characterize, and try to avoid, Internet paths through surveillance states. None publishes a measurement of how much surveillance ordinary trips pass or what avoiding it costs; the routing engine here is the instrument for that measurement, not its contribution.

What is new here. To our knowledge this paper is the first to measure, nationally and on real commute flows, how many ALPR fields of view commuters pass and what avoiding them costs; the first to reconstruct the crowdsourced ALPR map from OpenStreetMap's edit record at monthly

resolution and follow a fixed sample of trips across it; and the first to check that map's coverage against the camera counts police departments publish.

3 The instrument

The measurement needs a router that treats "avoid these 159,286 polygons" as a cost rather than a post-hoc filter. GraphHopper 11.0 [20] supplies it without a fork: it resolves a directory of custom areas into a spatial index *at graph import time*, and—the documentation is ambiguous, so we verified it—an area registered that way can be referenced by name from a *per-request* custom model:

```
{ "priority": [ { "if": "in_alpr", "multiply_by": 0.01 } ] }
```

Shipping 159,286 polygons in every request is not viable; naming them is. The cost is that a new cone set needs a full re-import, about two hours for North America. Two settings follow. Contraction Hierarchies [16] are disabled, since they bake edge weights into the prepared graph and a per-request avoidance strength needs them mutable; landmarks, the ALT algorithm of Goldberg and Harrelson [18], serve instead. And the avoiding profile shares the plain profile's landmark preparation, which is valid only because avoidance multiplies priority strictly downward, so the landmark distances remain the lower bounds ALT requires. The map's cone geometry also serves TRAMES, a fork of the OsmAnd navigation client [37] that routes around the cameras on the phone itself: its offline maps tag every road a cone touches, and OsmAnd's own router penalises those roads with the same multipliers. A public instance of this engine served the app's optional online mode until 29 August 2026; the study's routing server binds to localhost and is not internet-facing. The app avoids the volunteer map alone: the registry's readers that this paper's current figures count (Section 4.2) are not in it, and Section 7 measures what that leaves on a route planned around the map alone.

4 Data

All sources are public and key-free.

4.1 ALPR installations

Source: OpenStreetMap [34], queried through the Overpass API [33] for nodes tagged `man_made=surveillance` with `surveillance:type=ALPR`, matched case-insensitively (19 readers are tagged `alpr`)—the corpus surfaced by the DeFlock project, taken from the upstream rather than a mirror. Retrieved in five regional queries (continental US, Alaska, Hawaii, Canada, Mexico) and deduplicated by OSM node id. The map in this paper is the snapshot of **2026-09-22: 142,991 unique camera nodes**, 142,257 of them in the 50 states and the District of Columbia; the rest fall in the parts of Canada and Mexico the regional queries cover. With the registry's readers (Section 4.2) the merged map holds 186,316 installations in the states and the District (Figure 1). A first snapshot, taken 2026-07-24, held 120,838 (120,272 in the states), and Section 4.4 reconstructs the map on the first of every month since January 2024.

A fetch that fails over between Overpass mirrors has to check the mirror's database date: on the first attempt one regional query was answered from a mirror four months stale, and roughly 13,600 cameras silently reverted to old tags with nothing reported as failed. The fetch now rejects any answer older than 48 hours, and every tile of the snapshot used here falls within a 14-minute window.

Table 1: Directional and vendor tagging across the two 2026 snapshots. The direction-key rows overlap (392 and 415 nodes carry both keys). Vendors are matched as case-insensitive substrings of the first of the four vendor keys present, in both columns.

	2026-07-24		2026-09-22	
Property	Count	Share	Count	Share
Camera nodes	120,838		142,991	
Any direction tag	119,249	98.7 %	140,112	98.0 %
via `direction`	117,356	97.1 %	138,216	96.7 %
via `camera:direction`	2,285	1.9 %	2,311	1.6 %
Multi-head (`;`-separated)	7,014	5.8 %	9,150	6.4 %
Arc-range form (`199-269`)	4,806 tokens	—	4,632 tokens	—
`operator` tagged	22,354	18.5 %	23,872	16.7 %
Flock Safety	99,650	82.5 %	114,360	80.0 %
Motorola Solutions	5,993	5.0 %	7,411	5.2 %
Genetec	2,876	2.4 %	3,721	2.6 %
Leonardo	1,020	0.8 %	1,215	0.8 %
Axis Communications	1,052	0.9 %	3,327	2.3 %
No vendor tag	5,811	4.8 %	6,850	4.8 %

Three tagging realities shaped the parser. Mappers write the bearing under plain `direction`, not the `camera:direction` the wiki documents [36], by a factor of 60; reading only the documented key would discard 98 % of the bearings. Explicit arcs are real and mostly 45°: 4,632 range values on 4,372 nodes, 3,806 of them (82 %) spanning exactly 45°, which is what justifies 45° as the default span for the 97 % of cameras that give a bare bearing—the default is inherited from the data, not assumed. And multi-head units exist (`320;190`, `0;72;144;216;288`): 9,150 nodes, typically watching both carriageways, so the parser emits one wedge per head rather than silently taking the first, which would make a camera that sees everyone look like one that sees half.

4.2 Flock Safety's device registry

For the present, the volunteer map can be supplemented with the vendor's own records. An independent researcher has published a map of Flock Safety's device registry as it stood in December 2025 [29, 30], which we retrieved on 25 September 2026. Its 335,701 records, 334,525 of them in the fifty states and the District, give each device's position, type (a *Falcon* reads plates; other types record video or sound), status (in service, planned or decommissioned), a label that often names the road and the traffic watched (`#02 Pacific Pointe (Outbound)`), the date its record was created, and a `rotationAngle`, which is not a bearing: on the 73,226 paired devices whose mapped node carries one surveyed bearing, the two disagree by a median of 90°, as two independent angles would, and agree within 20° for 7.8 %. No direction is derived from it.

Each registry device is paired with at most one mapped node, and each node with as many devices as it has heads, within 50 m, nearest first, the two sides' vendor and capability required to agree: 88,754 pairs, covering 63.8 % of the registry's 126,389 in-service plate readers, at a median separation of 5.6 m. 54,372 mapped nodes have no registry device within reach—another vendor's camera, or one the registry does not carry—and 45,762 in-service plate readers in the registry have no mapped node. We drop 1,586 of those whose records place them at a coordinate shared by many devices (a building's address rather than a pole, 1,473), name them as Flock's own or test units, say

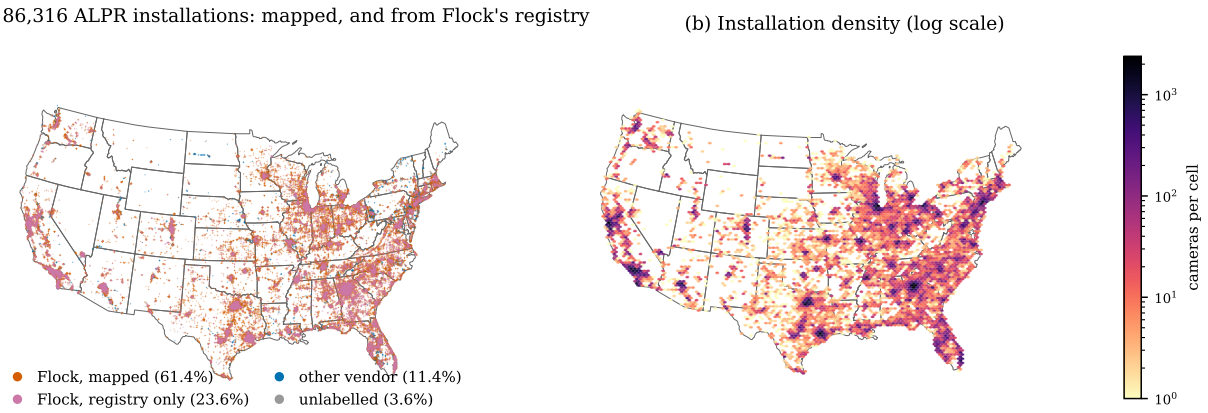


Figure 1: ALPR installations in the contiguous United States: those mapped in OpenStreetMap on 22 September 2026, and the in-service readers in Flock Safety's December 2025 registry that the map lacks. (a) By source and vendor, the registry's readers drawn on top: they show where the volunteer map falls short. (b) Density of both, on a log scale. The visible structure is partly deployment and partly *recording*: see Sections 6.3 and 8.2. Camera data © OpenStreetMap contributors and Michael [30].

they are retired, or duplicate another record, and add the remaining **44,176** to the map: 44,059 in the fifty states and the District, 117 in Puerto Rico and the Virgin Islands, where no sampled commute runs. Planned and decommissioned readers, and devices that do not read plates, stay out. The registry adds only Flock's devices, so on the merged map Flock's share of installations rises from 80 % to 85 %; other makers' unmapped cameras remain unmapped.

These are not recent cameras the map has yet to reach: 26,902 of the 44,176 carry the registry's bulk-import date of 26 March 2024, so were on record by then, and only 2,150 were recorded in the second half of 2025. A record's creation date is not an installation date—a device is recorded while planned—and the registry gives no date of removal, so it cannot be reconstructed as it stood on an earlier day. The trend of Section 6.4 therefore rests on the map alone.

4.3 Cone construction

Each camera head becomes a circular sector: apex at the node, bisector along the tagged bearing, span 45° (or the tagged arc), radius 60 m. The radius must reach across the carriageway from a set-back pole; it was chosen on that reasoning and then swept over $\{30, 45, 60, 90\}$ m (Section 8.4). Measured exposure is only mildly sensitive to it, but the avoidance result is not robust above the assumed value.

All wedges are unioned into a single MultiPolygon of **135,210 parts**, published as one feature with id `alpr`. Overlapping wedges merging is correct and intended: the question is *is this road segment observed*, not *by how many devices*.

A camera with no bearing receives no cone and contributes no exposure: 2,879 nodes carry no direction tag and 69 an unparseable one, 2,948 of 142,991 (2.1 %). A full circle would overstate a camera that watches one approach; excluding them understates by at most that share, and makes measured exposure a lower bound on the map's own terms as well as on the world's. Thirty-one nodes give a full-circle arc (`0-360`); in this run the parser gave them a 45° wedge, an error since corrected to a disc of the same radius and too small—31 of 140,043 cameras with cones—to move any figure reported here. The vendor is read from four inconsistently spelled keys by case-insensitive

substring (114,360 Flock cameras, against 109,892 by exact match); it matters only for the vendor split of Section 6.1, and the avoidance model targets the all-vendor union throughout.

The registry's readers carry no bearing at all, and excluding 44,176 cameras would discard the registry. Each receives instead a **disc of the cone's radius**, 60 m in every direction, the rule the published device map applies to any device of unknown heading. A disc watches both carriageways where a Falcon watches one, so it can overstate a reader's view; Section 5.2 scores the discs a second way, from the direction named in the device's label, and Section 6.1 reports both. The merged set is the map's wedges, rebuilt with the corrected parser (135,202 parts), plus the discs: **159,286 parts**, one feature with id `alpr_merged`, imported as a graph of its own.

4.4 Camera maps of earlier dates

OpenStreetMap keeps every edit, so the camera map can be reconstructed as it stood at any past instant. The public Overpass server answers a query carrying a `[date:...]` qualifier from its history: nodes deleted since are restored, nodes added since are absent, and every tag has its value from that day. Asked for Georgia at the July snapshot's timestamp it returned 9,742 nodes, the snapshot's own count to the node. History queries for the whole continent time out on the public server, so a date is fetched state by state and merged by node identifier; DeFlock's own Overpass mirror keeps no history and cannot serve them. They are heavy, and after one complete date the public server began refusing our connections.

Every other date therefore comes from OpenStreetMap's full-history dump [35] (163 GB, 14 September 2026, checked against its published MD5), read once. Two passes collect every version of every node that was ever tagged as an ALPR—169,565 nodes and 248,769 versions, including the later versions that deleted or retagged them—and the map at any instant is then a filter over that file, with Overpass's semantics: each node's latest version at or before the instant, kept if it was visible and still tagged. On the one date both sources cover, 1 July 2025, they agree exactly—22,690 nodes, the same identifiers, the same tags, the same coordinates—and on 1 January 2026 the filter reproduces an earlier, independent pass over the dump node for node (63,526). The same file gives the day, to the second, on which each camera entered the map, and a monthly count of the map that needs no aggregating service.

We reconstruct the map on the first of every month from January 2024 to September 2026, thirty-three maps, and set them beside the study's snapshots of 24 July and 22 September 2026. Cones are built from each exactly as from the current map (Section 4.3). The share of mapped cameras carrying a bearing, and so receiving a cone, was about 90 % in 2024 and 98 % in 2026: the earliest maps are slightly less complete even on their own terms. Exposure is measured on every monthly map; avoidance is routed on the first of every quarter (Section 5.3).

One caveat governs everything built on these maps. OpenStreetMap records when a camera was *mapped*, not when it was installed: an installation date is present on 47 of 142,991 nodes. A camera first mapped in 2026 may have been on its pole since 2021. A past map is therefore what a study run on that day would have seen, and the trend of Section 6.4 describes how the measured picture changed as volunteers filled in the map—most of it after DeFlock was founded in October 2024 [4]—not how fast cameras were deployed.

4.5 Road network

Geofabrik North America extract [17], 19.2 GB, retrieved 2026-07-24 and held fixed for every camera map, so that the trend of Section 6.4 isolates the camera data: every unavoided route is the same route on every date. Imported with `ignored_highways = footway,construction,`

`cycleway,path,steps`. `track` is deliberately retained: unpaved tracks are plausible avoidance routes, and excluding them would bias the cost estimate downward by removing options a determined driver has.

4.6 Commute sample

Source: LEHD **LODES8** [42], file `od_main_JT00`: all jobs with both ends in-state, built from unemployment-insurance wage records linked to employer establishments. These are administrative counts of real jobs, not survey estimates. The year is 2022 for 49 of the 51 jurisdictions; LODES has not published 2022 for Michigan, which uses 2021, or for Alaska, whose latest year is 2016. LODES8 codes every year against 2020 census geography, so an older year still names current tracts.

Sampling design carries more weight here than sample size. Uniform random points would over-represent empty rural space where nobody commutes and no cameras exist, understating exposure badly. Instead each origin–destination pair is drawn **with probability proportional to the number of workers actually making that trip**, so the distribution approximates what an average *commuter* experiences rather than what an average *square kilometre* contains.

Block-level LODES geocodes are aggregated to **tract** (first 11 digits), finer than the routing's effective precision, and each tract is routed from its 2024 Census Gazetteer internal point [41] (`INTPTLAT/INTPTLONG`), which is guaranteed to fall inside the tract polygon. Same-tract commutes are dropped: origin and destination centroids coincide, so there is no route to measure. Georgia alone yields 1,199,250 distinct tract pairs covering 4,190,313 workers; across the 50 states and the District of Columbia the frame holds 132,822,679 commuters over 34,420,305 tract pairs.

Connecticut needed one repair: it replaced its counties with planning regions in 2022 and every tract identifier changed, so LODES and the Gazetteer name the same tracts differently and the state would silently drop out; a crosswalk on the six-digit tract code matches all 879 tracts in the flows.

1,200 pairs are drawn per jurisdiction, 61,200 in all, with replacement from a single seeded stream (seed `20260725`), giving **58,421 distinct commutes**. The fifteen states of the study's earlier rounds are drawn first, so their 17,904 commutes are reproduced exactly and their routes carry over. The sampled pairs represent 2,842,747 workers; the median pair carries 16 and the largest 2,832.

A pair drawn more than once is routed once, since its route is the same, but it stands for every draw: **each commute carries its draw count as a weight**. In the fifteen large states this is immaterial—0.53 % of draws are repeats—but in small ones it is not. Wyoming's 1,200 draws hold 832 distinct pairs (30.7 % repeats), Alaska's repeat a quarter of the time and the Dakotas' nearly a fifth, and one Vermont pair was drawn nine times. Collapsing repeats at equal weight, as this study's earlier rounds did, under-represents precisely the heaviest flows; keeping the counts restores probability-proportional-to-workers sampling inside every state.

The pairs are what LODES records, not trips anyone has been seen to make. The median routed commute is 23.6 km, but one pair in eight is more than 100 km apart by road (12.8 % of commuters, weighted) and one commuter in a hundred more than 470 km: LODES places each job at an employer establishment, imputed where an employer has several, and pairs that far apart are unlikely to be driven daily. They are kept, since any cut-off would be arbitrary, and their weight is reported where it matters (Sections 6.1 and 8.6).

5 Method

5.1 Two routes per commute

Only the trip to work is routed. A camera watches one direction of travel, so the journey home passes a different set of cameras; it is not measured here, and no figure in this paper should be doubled to stand for a day. Each sampled pair is routed twice against the same graph: a **baseline** with no custom model, and an **avoided** route applying `priority: [{if: "in_alpr", multiply_by: 0.01}]`. Both disable CH, which per-request custom models require.

Both use GraphHopper's stock car model, which weighs distance as well as time (`distance_influence` 90: a route 1 km shorter is preferred if it costs under 90 s more). The baseline is therefore the router's preferred route rather than strictly the fastest, and for 3.9 % of commuters on the merged map (4.5 % on the map alone) the avoiding route—in every such case the longer one—is the quicker, by a median of 0.9 min. Extra time is reported as measured, negative values included; clipping them at zero leaves every median unchanged and raises the mean extra time from 7.39 min to 7.45 min (on the map alone, from 5.62 min to 5.70 min).

0.01 is the application's *strongest* avoidance setting, not a typical one: a watched road costs a hundred times its usual weight, close to but short of prohibition. Results therefore describe the **strong end of both the benefit and the cost** of avoidance as offered. This is deliberate: it establishes whether meaningful avoidance is possible at all, and what it costs near the limit.

5.2 Exposure is directional, not proximate

A camera counts as having seen the driver *only if the route line intersects its cone*. Passing 20 m from a camera aimed the other way is not exposure. Implementation is an STRtree over the cone parts for candidate lookup, then an exact intersection test.

Two simplifications remain, and they pull in opposite directions: route geometry is a polyline while a vehicle has width, which understates exposure, and a cone occluded by intervening buildings is still counted as full sight-line, which overstates it.

A registry disc has no direction, so a route that crosses one counts it. The device labels supply a second, directional, reading: 26,131 of the 44,176 name a cardinal direction of travel (*northbound*, ...) and 4,161 say inbound or outbound. Under the *directed* scoring a cardinal disc counts only if the route crosses it travelling within 60° of the named heading, taken from the route's bearing where it passes nearest the device; the other discs count as before. The two scorings bracket the registry's contribution: every reader on both carriageways, or every reader on the one its label names.

5.3 Earlier camera maps need no new baselines

The unavoided route does not depend on the cameras. Routed two months apart against graphs carrying different camera maps, all 17,580 commutes of the study's earlier rounds took identical baseline routes, to the metre and the second. Exposure on an earlier date is therefore a re-scoring of each saved route against that date's cones, with no routing. The merged map is treated the same way: the saved baselines re-scored against its cone set, and avoidance routed against a graph carrying that set for every commute the merged map exposes.

Avoidance does need routing against that date's cameras, and two properties keep it tractable. First, one graph serves every date: each date's cone set is imported as its own custom area (`alpr_2025_07_01`, ...), and each request names the one it avoids —an import of about two

hours serves five or six dates rather than one, checked on a test graph to route each area independently. Second, only commutes that pass a camera on that date are routed. A route that crosses no cone is already the avoiding route, since the custom model only makes cone edges dearer and nothing can undercut it; this was checked rather than assumed, and all 3,741 such commutes of the fifteen-state run received an identical avoiding route. The same shortcut serves the main run.

Re-scoring uses route geometry stored at five decimal places, about a metre. Against live scoring on the current map it moves 1.7 % of commutes by one camera, in the two directions about equally, changes whether 5 of 17,580 (0.03 %) pass any camera at all, and leaves national totals within 0.1 %. On the full sample the re-scored national figures match the live ones to the precision reported (75.3 % passing a camera, a mean of 5.14). Within the trend every date, the current one included, is scored the same way. Exposure is re-scored on every monthly map; avoidance is routed on the first of every quarter, eleven dates from January 2024 to July 2026, and on the September 2026 snapshot itself. The 24 July 2026 snapshot enters re-scored only; the fifteen-state paired comparison covers it (Section 6.4).

5.4 Statistical reporting

- **Medians lead, means accompany.** Both distributions are heavily right-skewed; a minority of long commutes through dense corridors pull the mean well above the typical experience.
- **National figures are weighted by commuters.** Every jurisdiction was sampled at the same size, so an unweighted pool would count Wyoming's 198,447 in-state commuters as heavily as California's 16,821,903. Each routed commute carries its state's commuter total from the LODES frame, shared among that state's commutes in proportion to draw count; medians are weighted quantiles, and state figures need only the draw counts. Weighting alone moves the share of commuters passing a reader in the fifteen states of the earlier rounds from 82.5 % to 84.3 % on the merged map (from 78.7 % to 81.1 % on the map alone), because the populous states are the more surveilled ones.
- **Intervals from a bootstrap stratified by state** [8]: percentile, 2,000 replicates, each resampling every state at its own size and re-deriving its weights, as the sample was drawn. Commutes from one county share corridors and cameras; resampling whole origin counties within states instead [13] widens the headline intervals two- to three-and-a-half-fold—78.6 % becomes [77.2–79.4], the median 2.29 min [2.12–2.56]. Where the two disagree, both are shown.
- **Differences between the two maps are paired.** Both runs route the same commutes, and the bootstrap draws the same replicates for each, so a difference is bootstrapped replicate by replicate rather than read off two separate intervals.

5.5 Execution

Twelve concurrent workers against the local routing server, past which contention outweighs added concurrency. Failed routes are dropped rather than retried, and **56,131 of 58,421 commutes routed (3.9 % dropped)**. Nine in ten failures are points the router cannot snap to any road—a tract's internal point deep inside a national park, a reservation or a wilderness tract—and the rest are genuinely unconnected, such as commutes between Hawaiian islands. Five states lose more than a tenth of their sample (Alaska 33 %, Hawaii 24 %, Montana and Wyoming 21 %, New Mexico 14 %); the median state loses 2.0 %. Those five hold 1.6 % of the nation's commuters, so the national figures barely register the loss, but their own estimates describe the routable, probably less remote, part of each state. A wider road search could snap more of them, but in a test it moved 5 of 100

Table 2: Exposure and the cost of avoidance for American commuters, on the merged map (OpenStreetMap's cameras of 22 September 2026 with the in-service readers only Flock Safety's registry records) and on the map alone: 56,131 routed commutes in the 50 states and the District of Columbia, weighted to the 132.8 million in-state commuters they represent. Medians are weighted; intervals come from a bootstrap stratified by state. Median commute 23.6 km / 23.4 min.

	Merged map		Map alone	
	Median [95 % CI]	Mean	Median [95 % CI]	Mean
Unavoided route				
Cameras passed	3 [3–3]	6.24	3 [3–3]	5.14
Commuters passing ≥1	78.6 % [78.1–79.0]		75.3 % [74.8–75.8]	
Commuters passing ≥5		42.0 %		36.5 %
Commuters passing ≥10		20.7 %		16.3 %
90th / 99th percentile	16 / 42 cameras		13 / 34 cameras	
Exposure rate (per km)	0.117 cameras		0.096 cameras	
ALPR-avoiding route				
Cameras passed	0 [0–0]	0.36	0 [0–0]	0.26
Reduced to zero	**80.8** % [80.3–81.3]		84.2 % [83.7–84.7]	
Extra time (min)	**2.29** [2.21–2.37]	7.39	1.62 [1.57–1.68]	5.62
Extra distance (km)	0.87 [0.83–0.91]	3.37	0.57 [0.55–0.60]	2.53
Avoidance overhead	9.71 % [9.48–9.97]	17.22 %	6.71 % [6.50–6.93]	13.27 %
Minutes per camera evaded	0.83 [0.82–0.85]	1.38	0.76 [0.75–0.78]	1.30

already-routed commutes to a different road, and the trend of Section 6.4 pairs every saved route with new ones; consistency was kept over coverage. Commutes are processed in seeded random order, so any prefix of the output is a near-uniform sample of the whole. Against the merged map the 37,725 commutes it exposes were routed again, none failing once 72 interrupted requests were repeated; the two runs' unavoided routes are identical to the metre and the second, and the merged camera count re-scored from stored geometry matches the router's on every one.

6 Results

6.1 Exposure on the unavoided commute

78.6 % of American commuters pass at least one ALPR on the way to work. The median commuter passes three and the 90th percentile sixteen, and aggregate exposure is 0.117 cameras per kilometre driven (Table 2). The 99th percentile, forty-two, belongs mostly to the long pairs of Section 4.6: the commuters there travel a median 290 km one way, and among the 87 % whose pair lies within 100 km the 99th percentile is thirty-one. Against the map alone, the registry's readers add 1.09 encounters to the average commute [1.06–1.13], a fifth more, and 3.2 points [3.0–3.5] to the share passing any reader. They mostly thicken what the map already shows: the commuters the map alone exposes pass a median of five readers instead of four, while 3.3 % [3.1–3.5] pass one only once the registry is added.

A disc sees both carriageways, where a Falcon watches one. Scored instead by the direction each reader's label names (Section 5.2), 77.7 % [77.2–78.1] of commuters pass a reader and the mean is 5.82: 38 % of the encounters the registry adds are with readers whose labels say they face the other way. The share passing any reader, and the avoidance results, move by less than a point; scored

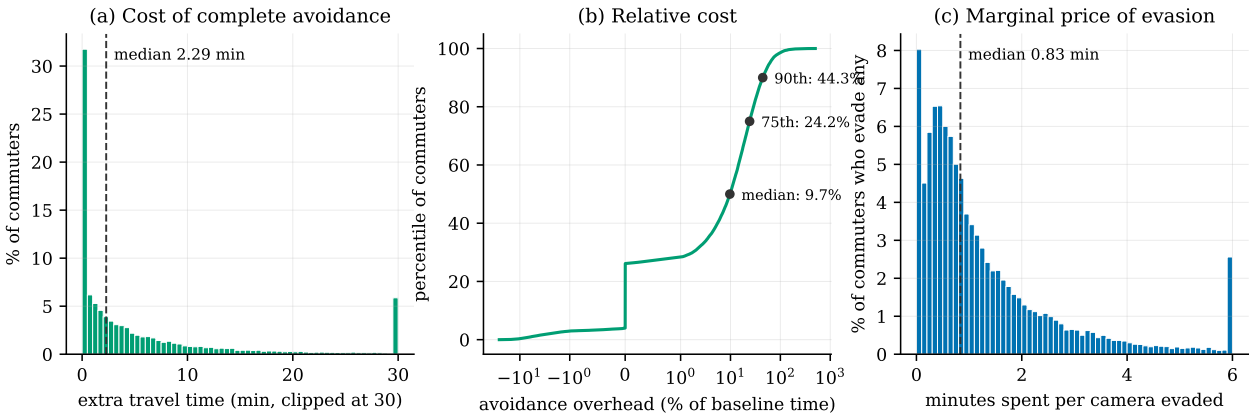


Figure 2: The price of complete avoidance, as shares of commuters. (a) Absolute extra time; (b) overhead as a percentage of baseline, by percentile; (c) marginal minutes per camera evaded. All three are heavily right-skewed—a small tail of commutes pays a great deal, while the typical commute pays almost nothing.

by direction, the avoiding routes reach zero for 81.1 % of commuters, against 80.3 % for the same stored routes scored with discs.

By vendor, on the map alone, where every cone comes from a surveyed bearing, Flock Safety holds 81.5 % of the installations that contribute cones in the fifty states and the District but supplies 67.8 % of commuter exposure; per unit deployed, other makers' cameras generate 2.26 times as many encounters, being multi-headed more than six times as often (23.3 % against 3.6 %) and standing more often beside principal arterials and freeways (42 % within 30 m of one in the federal traffic inventory [11], against 26 % of Flock's). Where the operator is tagged, police departments run about two-thirds of both. The registry adds only Flock's readers, and as discs, so a per-unit comparison on the merged map would measure the disc rather than the vendor; on that map Flock holds 85 % of all installations, against 80 % of the map's. Each commute's saved route re-scored against each vendor's cones gives the split without new routing; the avoidance model does not distinguish vendors.

6.2 The cost of refusal

80.8 % of American commuters could be routed to zero ALPR exposure: the fifth whose usual route passes no known reader, and three in four of the rest (75.5 % [74.9–76.1], Table 3). Over all commuters the avoiding route adds a median of 2.29 min and 0.87 km, an overhead of 9.71 %; for commuters whose usual route passes a camera it adds a median of 3.92 min [3.82–4.02]. The median marginal price is 0.83 min per camera evaded.

The registry's readers make refusal dearer on every measure. On the same commutes, against the map alone, the share able to reach zero falls by 3.5 points [3.1–3.8], the median extra time rises by 0.67 min [0.62–0.72], about two-fifths, and the price of evading a camera by 0.07 min [0.06–0.08]. For the commuters the map alone already exposes the median detour grows from 3.14 min to 4.12 min, and 4.5 % of all commuters [4.2–4.9] who could reach zero against the map cannot once the registry is added. Extra time rises for a third of commuters, is unchanged to within three seconds for three in five, and falls for one in twenty, where the router, finding its old detour watched too, more often keeps a camera (43 % of those routes do, against a quarter of all exposed commuters' routes).

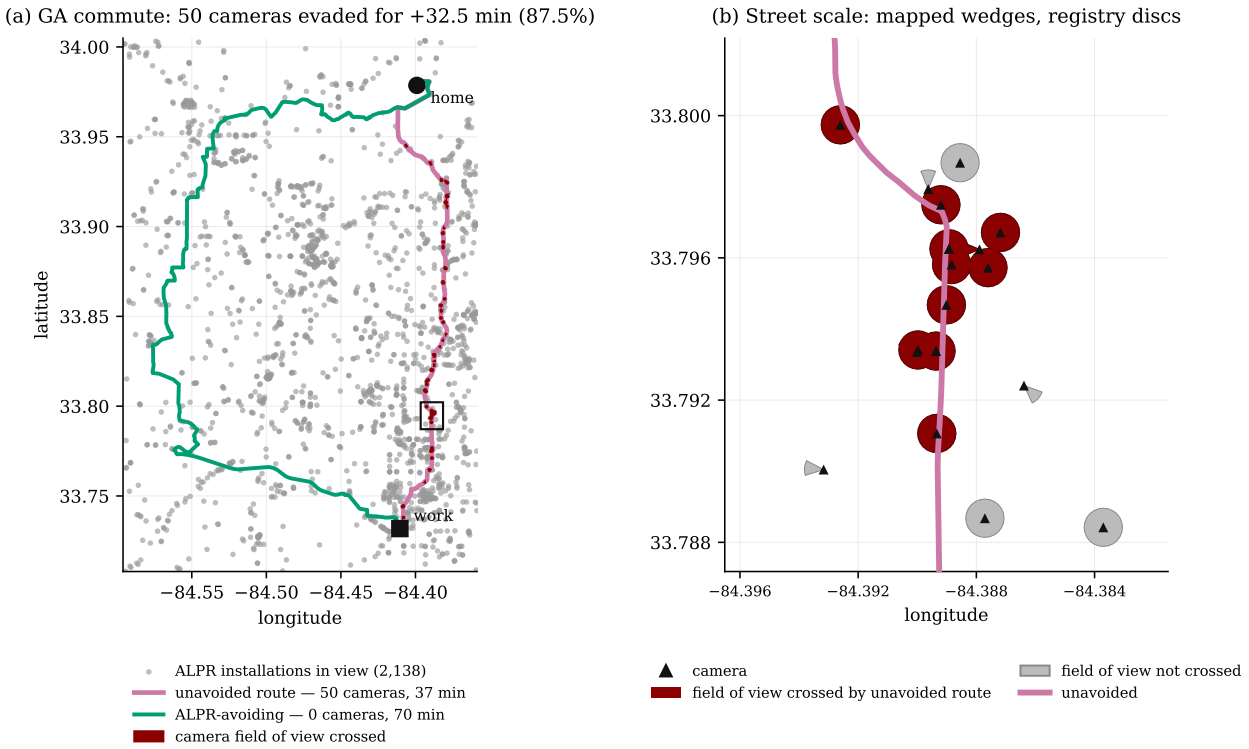


Figure 3: An illustrative high-exposure commute in metropolitan Atlanta—chosen for legibility, not typicality, and from the far tail of the cost distribution. (a) The unavoided route crosses 50 camera fields of view; the avoiding route crosses none, for 32.5 min (87.5 %). (b) At street scale, the mapped cameras are wedges aimed by their surveyed bearings and the registry's readers, whose direction is unknown, are discs; fields of view the unavoided route crosses are dark, those it misses grey, and they cost nothing. Map data © OpenStreetMap contributors.

For most commuters, then, refusing licence-plate surveillance on the way to work costs a few minutes at most. This is conditional on the assumed camera field of view: it is stable if the true radius is smaller than the 60 m assumed here and degrades if it is substantially larger, which Section 8.4 quantifies. It is the paper's central quantitative claim.

The distribution matters as much as the median. The mean overhead is 17.22 % against a median of 9.71 %, and the interquartile range runs from 0 % to 24.2 %: a quarter of commuters pay nothing at all, while a tail pays substantially. Figure 3 shows one such case.

6.3 Fifty states and the District of Columbia

State-level variation is extreme (Figure 4; every state's figures are tabulated in the repository). In Georgia 93.9 % of commuters pass a reader, a mean of 16.6 on the way to work; only 37.5 % can reach work past none, and avoiding costs a median of 13.6 min, an overhead of 44.4 %. In Illinois 91.6 % pass one, in California 89.4 %, in Florida 89.1 %. At the other end 14.8 % of New Hampshire's commuters pass one, 12.0 % of Montana's, 5.4 % of Vermont's, and none of Alaska's sampled commuters. In fifteen jurisdictions the median extra time is zero: for at least half of their commuters the avoiding route takes no longer than the usual one, in eleven of them because most commuters pass no reader at all. The registry moves some states far more than others, since it

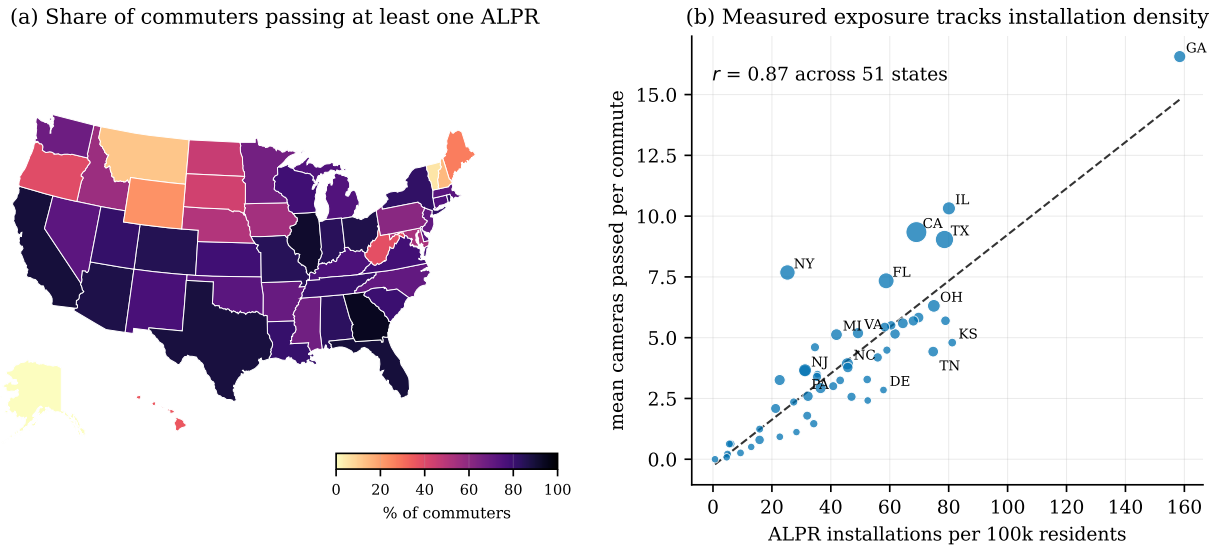


Figure 4: (a) Share of each state's commuters whose unavoided route passes at least one ALPR on the merged map. (b) Mean cameras passed against installations per 100,000 residents, points sized by commuters. Installation density alone tracks most of the variation between states ($r = 0.87$): from the record alone, a state with few cameras cannot be told from one whose cameras are unrecorded (Section 8.2).

adds only where Flock operates and mappers had not reached: Arizona's share rises from 72.7 % to 87.0 % and Massachusetts's and Wisconsin's by seven points, while New Hampshire, Vermont, Wyoming and Alaska do not move.

These are differences in the record at least as much as on the roads. Installation density runs from 0.7 per 100,000 residents in Alaska, and under five in Vermont and New Hampshire, to 158 in Georgia—an eight-fold spread between the 10th and 90th percentile states—and state exposure tracks it at $r = 0.87$. The departures from the line are what a statewide rate would produce: New York's commuters pass more cameras than its statewide density predicts, because commuting concentrates where cameras do while the density is averaged over the whole state; Kansas's pass fewer.

Whether evasion is cheaper per camera where cameras are denser, the data do not settle. On the map alone, across the 47 states and the District with at least 100 sampled commutes that evade a camera, the median price per camera falls weakly as cameras per kilometre rise, from 2.49 min in New Hampshire to 0.17 min in the District of Columbia ($r = -0.29$ against the logarithm of cameras per kilometre; a Fisher interval excludes zero, -0.53 to -0.01, but a bootstrap over states does not, -0.63 to 0.23). On the merged map the correlation is weaker still, $r = -0.19$, and neither interval excludes zero (-0.45 to 0.10; -0.58 to 0.33). Half the registry's readers stand in four states the map already showed as heavily watched—Georgia, California, Texas and Florida—and there the price of evading each rises: Georgia's from 1.11 min to 1.32 min. The extremes hold, New Hampshire dearest and the District cheapest, but a tendency between them is not one we can claim.

6.4 Thirty-three months of mapping

Everything above describes the present, on the merged map. The registry records neither when a reader went up nor when one came down, so the past can be measured on the map alone. We

Table 3: The same 56,131 commutes on the same roads against the camera map of each routed date, commuter-weighted. Exposure from re-scoring each saved route; avoidance from routing every commute exposed on that date, with the share reaching zero and the median extra time given for those commuters. The monthly maps between these dates are in Figure 5. The last row is the merged map: the September 2026 map with the registry's readers, routed in full, whose exposure columns come from the router rather than from re-scoring. Intervals come from the trend's own bootstrap and can differ from Table 2's in the last digit.

Map date	Mapped ALPR	Passing ≥1 (%)	[95% CI]	Mean passed	Exposed who reach zero (%)	Extra min, exposed	Min per camera
2024-01-01	1,112	2.2	[2.0–2.4]	0.05	95.0	0.77	0.42
2024-04-01	1,267	2.9	[2.7–3.1]	0.08	95.7	1.04	0.51
2024-07-01	1,420	3.3	[3.0–3.5]	0.08	96.0	1.00	0.49
2024-10-01	1,581	3.5	[3.2–3.7]	0.09	96.0	1.02	0.52
2025-01-01	7,583	12.3	[11.9–12.7]	0.35	92.5	1.20	0.61
2025-04-01	14,238	20.4	[19.9–20.9]	0.59	92.7	1.24	0.63
2025-07-01	22,690	29.4	[28.8–30.0]	0.94	92.4	1.36	0.65
2025-10-01	34,383	38.4	[37.8–39.0]	1.33	91.6	1.51	0.66
2026-01-01	63,522	55.9	[55.3–56.4]	2.44	86.9	1.98	0.70
2026-04-01	85,845	63.4	[62.8–63.9]	3.14	84.7	2.32	0.73
2026-07-01	109,853	69.6	[69.1–70.1]	4.08	82.6	2.67	0.75
2026-09-22	142,257	75.3	[74.8–75.7]	5.14	79.0	3.14	0.76
+ registry	186,316	78.6	[78.1–79.0]	6.24	75.5	3.92	0.83

re-asked the same questions of OpenStreetMap's camera map as it stood on the first of every month from January 2024, reconstructed exactly from its history (Section 4.4), and of the July 2026 snapshot: the same 56,131 commutes on the same roads, each saved route re-scored against each month's cones, and, on the first of every quarter, every commute that passed a camera on that date routed again to avoid that date's cameras (Section 5.3). Table 3 gives the routed dates, Figure 5 every month, and Figure 6 every state; the last row of the table and the hollow markers of the figure set the merged present beside the map's. Each date should be read as what had been *mapped* by then, not what had been installed.

The map filled in. On 1 January 2024 OpenStreetMap held 1,112 ALPR installations in the fifty states and the District of Columbia, four-fifths of them with no vendor tag, and twenty-six jurisdictions had none at all. It grew by about fifty a month until DeFlock was founded in October 2024, and then abruptly: 1,724 on 1 November, 4,894 on 1 December, 7,583 on 1 January 2025. From there the map grew by a median 14.1 % a month—22,690 by July 2025, 63,522 by January 2026, 109,853 by July, and 142,257 on 22 September 2026, a 128-fold increase in thirty-three months (Figure 5a). As a rate the pace has eased, to a median 9.2 % a month over the first eight months of 2026; in cameras it has quickened, from about 4,700 a month during 2025 to 9,200 in 2026. By July 2025 only Hawaii, Vermont and Alaska had no mapped reader; by January 2026 only Alaska; on 1 July 2026 none, though Alaska had none again on 24 July and 1 September. Flock's share of the mapped network went from 17 % to 71 % during 2024 and has stayed near four-fifths since, while the untagged share fell from 81 % to 5 % and the named non-Flock vendors rose from 1 % to 15 % as contributors tagged them.

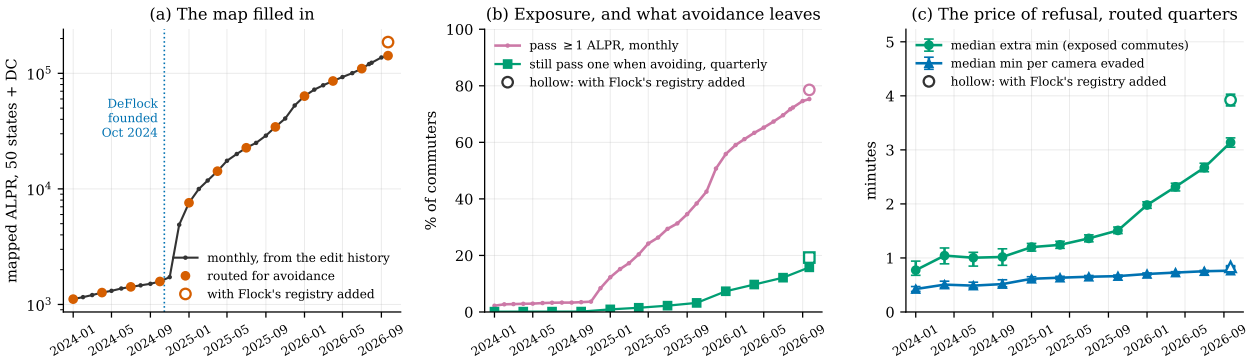


Figure 5: (a) Mapped ALPR in the fifty states and DC on the first of every month, from the edit history, with the routed dates marked. (b) Share of commuters passing at least one mapped reader on each monthly map (band: 95 % CI), and the share still passing one on the avoiding route, on the routed dates. (c) For commuters exposed on each routed date, the median extra minutes to avoid every camera, and the median minutes per camera evaded. Hollow markers: the September 2026 map with Flock Safety's registry added.

Exposure followed the map, almost in proportion. The share of commuters passing at least one mapped reader crossed a tenth in January 2025, a quarter in June, a half in December, and stands at three-quarters (75.3 %) today; the mean went from 0.05 cameras to 5.14, and adding the registry takes them to 78.6 % and 6.24. Against the number of mapped cameras the elasticity is 0.89 across the thirty-five maps. Per camera mapped, exposure has fallen by about a quarter since January 2024, from 0.049 to 0.036 encounters per commute per thousand cameras: the cameras mapped later lie on commute routes slightly less often than those mapped first, but not much less. Nor is there a sign of saturation. The two months between the July and September 2026 snapshots alone brought 18.3 % more mapped cameras and 17.0 % more encounters, and the monthly curve of Figure 5b is still rising by about two points a month.

The states filled in at different times. In January 2024 Illinois was the only state where more than a tenth of commuters passed a mapped reader (23.5 %); by January 2025 fifteen were, by July 2025 thirty-one, by January 2026 forty-seven, and today forty-nine (Figure 6). In July 2025 Michigan's commuters were the most exposed of any state's (62.8 %), while Pennsylvania (2.4 %), Nevada (3.8 %), New Jersey (4.4 %) and North Carolina (8.2 %) would have looked nearly camera-free; all four are between 60 % and 68 % today. A state's rank on the January 2025 map predicts its rank today imperfectly (Spearman $\rho = 0.80$ on mean exposure; from January 2024, when 37 jurisdictions had none, 0.54; from January 2026, 0.90). Nor does the map only grow. Wisconsin and Iowa each gained cameras on net between July and September 2026 yet lost about three points of exposure: 190 and 123 of their July cameras left the map, deleted or retagged, and the cameras newly mapped added less exposure than was lost.

Refusal got dearer, and not only because there was more to refuse. On the sparse maps of 2024 a commuter who passed a mapped camera could almost always get around it: on the first, 95.0 % [92.7–96.9] of exposed commuters could reach zero, for a median of 0.77 min, and through 2024 the share held at 95 % to 96 % for about a minute. On the September 2026 map 79.0 % [78.4–79.7] can, for a median of 3.14 min, and with the vendor's registry added 75.5 % [74.9–76.1], for 3.92 min [3.82–4.02] (Table 3; Figure 5c). The registry moves the present about as far as the

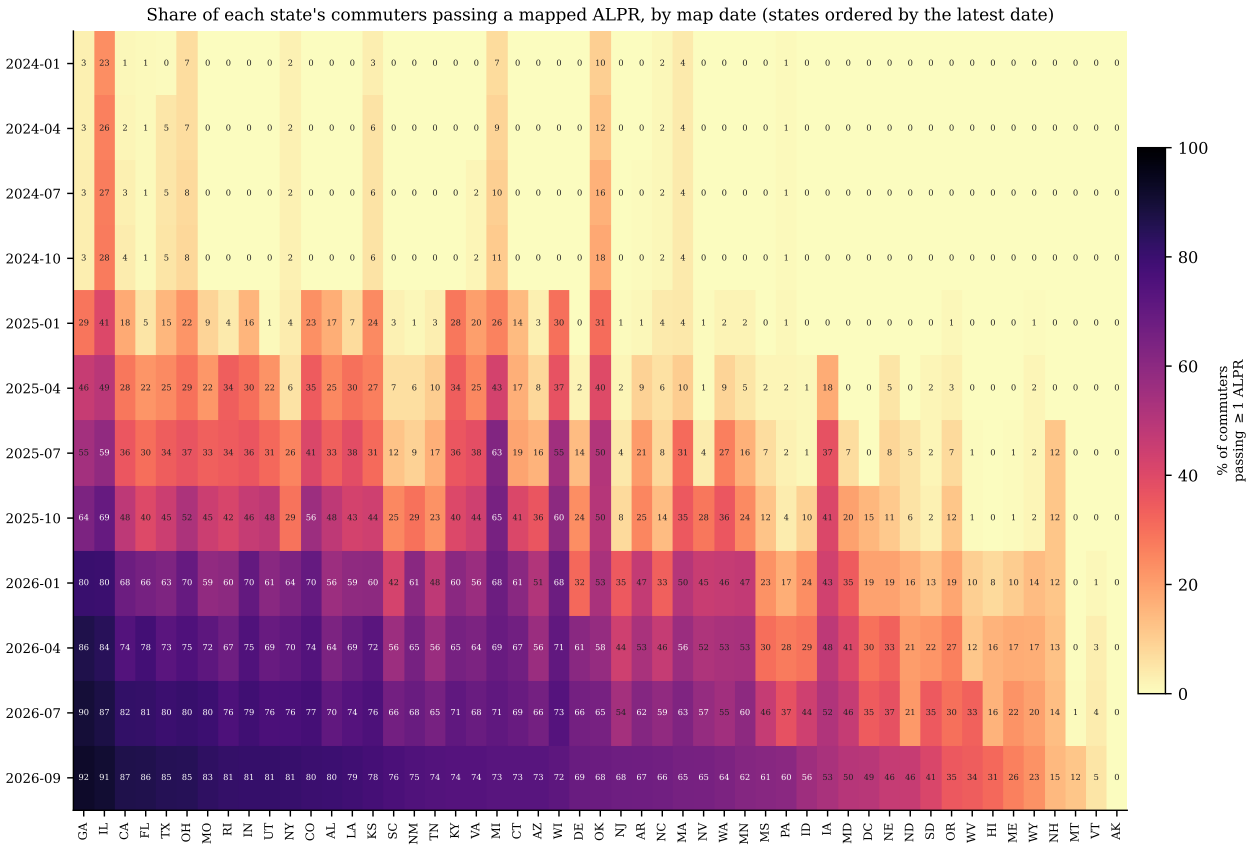


Figure 6: Share of each state's commuters passing at least one mapped ALPR on each routed map, states ordered by the latest.

preceding months of mapping did: the share reaching zero by as much as the twelve weeks from the July map, the median detour by as much as the nearly six months from April's. The quarterly maps show the shape of the change. The share reaching zero held at 95 % to 96 % through 2024, stepped down to about 92 % with the surge of mapping that followed DeFlock's founding and held there through 2025, and has fallen on every routed map since October 2025: 91.6 %, 86.9 %, 84.7 %, 82.6 % and 79.0 %, each interval clear of the last. The cost did not step; it climbed. From July 2024 on, the median extra time for an exposed commuter rose on every routed map, from 1.00 min to 3.14 min. Most of that rise is the number of cameras on each trip, but the price of evading each one rose too, from 0.42 min [0.38–0.47] to 0.76 min [0.75–0.78], and it too rose on every routed map from July 2024. Two things moved it, and a fixed sample can separate them. One is who was exposed: each fuller map reached commuters the sparser ones had not, on other roads with other alternatives. The other happened on the same trips. Followed from the July 2025 map to September 2026's, the 12,647 commutes exposed in July came to pass 9.9 cameras on average instead of 3.2; the share of them that can reach zero fell from 92.4 % to 72.6 %, and the price per camera rose from 0.65 min [0.63–0.68] to 0.81 min [0.79–0.84]. Every cohort shows it, and for about a year the rise grows with the time a cohort is followed: the commutes exposed on the July 2026 map pay 2 % more per camera twelve weeks later, too little to distinguish; April 2026's pay 7 % more, January's 11 % and October 2025's 20 %, each interval clear of its starting one. The seven cohorts followed for fourteen months or more pay 21 % to 35 % more, six of them clear of their starting intervals and the seventh, April 2024's, overlapping it slightly. As more of a trip's cameras, and of its alternatives', are mapped,

each costs more to avoid. Over the two months between the 2026 snapshots the change is too small to see, which is why the fifteen-state comparison below finds the price per camera unchanged.

Fifteen states, fully re-routed. The trend's last step can be checked against a comparison that re-routed everything. For the fifteen states of the study's earlier rounds, the same 17,580 commutes were routed in full against both 2026 snapshots on the same road extract, and every baseline route was identical to the metre and the second. Camera encounters grew by 18.3 % against 18.7 % more mapped cameras in those states, the median price of evading one stayed at 0.78 min to the second decimal, and Pennsylvania's mean exposure rose 88.5 % in two months.

What follows. Two things. Absolute exposure figures from a volunteer map are a lower bound that has not fallen from one month's map to the next and will keep rising; one vendor's records alone raise them again, exposure by a fifth and the price of refusal by two-fifths. Any study built on crowdsourced surveillance data dates, and should say when its map was drawn. And the price of refusal dates as well: it has risen on every routed map since July 2024, per camera as well as in total, and today's figures describe a record that is still filling in.

7 Discussion

7.1 Refusal is cheap, and that is the policy-relevant fact

The strongest result here is also the simplest: 80.8 % of American commuters can reach work past no known licence-plate reader, and the avoiding route is a median of 2.29 min longer—3.92 min for commuters with a camera to avoid. A common intuition about these networks is that they are, or soon will be, comprehensive enough that avoidance is futile, so that the privacy loss is unavoidable rather than chosen. At current density that is false for four commuters in five. It has been false on every map since 2024, but it is becoming less so: as the map filled in, the share of exposed commuters who can reach zero fell from 95 % to 79 %, and to 75 % once the vendor's registry is added, while the median cost for those with a camera to avoid quadrupled on the map alone and rises by a further quarter with the registry (Section 6.4).

Cheap refusal also presumes a router that knows where the cameras are. The app built on this study's cone geometry avoids the volunteer map alone (Section 3). Routes planned around the map alone are clean of it for 84.2 % of commuters, but for 35.0 % [34.4–35.7] of them the route still passes a reader that only the vendor's registry records, and for 44.1 % of those whose usual route passes a mapped camera; 54.7 % of commuters get a route clean of both. Avoidance is only as complete as the record it avoids.

The corollary cuts the other way. If avoidance is this cheap, it is also cheap for people with reason to evade legitimate investigation, which bounds how much investigative value a fixed-camera network can deliver against a motivated subject while it continues to read every unmotivated one.

7.2 Where refusal stops being cheap

The national figures average over states whose commuters have almost nothing to avoid and one where avoidance has largely stopped working. In Georgia, the most densely recorded state, 93.9 % of commuters pass a reader, the average commute passes 16.6, and only 37.5 % can reach work past none; avoiding costs a median of 13.6 min, an overhead of 44 %. On the map alone Georgia's commuters could reach zero in 54.3 % of cases, for 7.8 min: the 7,031 readers the vendor's registry adds there turn refusal into something most of them cannot complete. Each camera there also costs

more to avoid than nationally, 1.32 min against 0.83 min, which is the cohort finding of Section 6.4 in one place: the more of a trip's cameras, and of its alternatives', are on the record, the dearer each is to evade.

On the map alone the opposite pattern appears: evasion looks cheapest per camera where cameras are densest, a pattern that would rest on road-network structure, which OpenStreetMap maps close to completely, and so might be expected to survive the map's gaps in coverage. The vendor's registry is a test of exactly that, and the pattern does not survive it: across states it weakens to a correlation neither interval distinguishes from zero (Section 6.3). Which roads are watched is itself a question of coverage, so even a result built on the road network moves when the record of cameras does.

8 Limitations

8.1 Tract centroids are not addresses

Every route runs centroid to centroid; real commutes begin in a driveway. In large or irregular tracts the centroid can sit far from any residence. This adds noise in both directions and is the main reason we report distributions rather than point estimates. It also costs coverage: where a tract's internal point lies too far from any road to snap to, the commute cannot be routed at all, which removes a fifth or more of the sample in Alaska, Hawaii, Montana and Wyoming (Section 5.5).

8.2 Crowdsourced camera data is non-randomly missing

Absolute exposure figures remain a **lower bound**, now a tighter one. Like volunteered geographic data generally [21, 22], the map is least complete where contributors are absent, and there cameras are missing from the map but present on the pole. Against Flock Safety's own transparency portals the map alone holds, at the median, 0.98 of the cameras that 1,050 municipal police departments report operating (28,349 mapped inside their limits against 31,210 reported), small towns being the least completely mapped [10]; against the vendor's registry it holds 63.8 % of in-service plate readers (Section 4.2). The merged map closes that gap for one vendor, as of December 2025. What it cannot close: other makers' unmapped cameras, which the registry does not cover; Flock devices installed since December 2025, which the registry predates and the map holds only where mappers have reached them; and whatever the registry itself omits—Flock has said it operates more than 120,000 cameras nationwide [29], against the 126,389 in-service plate readers the registry carries, so that gap at least appears small. The trend of Section 6.4 is the best evidence of how far the map itself still has to go.

8.3 One vendor's registry, as it stood in December 2025

The registry closes the map's largest gap for one vendor at one moment, and inherits both limits. Its positions are the vendor's records rather than a survey: where a mapped node stands on the same device the two agree to a median of 5.6 m, but nothing checks the 44,176 readers no mapper has reached. Its statuses are December 2025's, nine months before the map's snapshot: a reader removed since is still counted, and the 34,397 the registry then listed as planned, many presumably installed since, are not. Its readers carry no bearing, so each is a disc rather than a wedge, which the directed scoring of Section 5.2 brackets. The merged figures can therefore err either way: the discs and any removed readers overstate, while the planned readers installed since and other makers' unmapped cameras (Section 8.2) understate.

8.4 Cone geometry: swept, and one result depends on it

The 60 m radius was chosen by reasoning—a pole set back from the kerb needs a cone long enough to reach the carriageway it watches—and the 45° default span from the 82 % of explicit arcs that span exactly that. Neither was measured. We therefore swept the radius over $\{30, 45, 60, 90\}$ m, rebuilding the map's cone geometry at each and re-scoring all 56,131 routes of the map-alone run (Table 4).

Table 4: Cone-radius sensitivity, commuter-weighted. Exposure is mildly sensitive to the assumption; the avoidance result is not, until the radius grows by half. At the assumed 60 m the clean share is 83.7 % rather than the map-alone 84.2 % of Table 2 because the sweep re-scores stored route geometry (Section 5.3).

Radius	Mean cameras	vs 60 m	Avoided routes still clean	Residual
30 m	3.82	0.74×	87.4 %	5.4 %
45 m	4.62	0.90×	85.6 %	5.2 %
60 m	5.14	assumed	83.7 %	5.3 %
90 m	5.88	1.14×	**55.1** %	21.7 %

Two things follow, and they point in opposite directions. Exposure is robust: halving the radius removes a quarter of it and 90 m adds 14 %, so Section 6.1 does not rest on the choice. The avoidance headline is conditional on it, and from above only: routes planned against 60 m cones stay clean when re-scored at 45 m and 30 m, but at 90 m the share reaching zero falls from 83.7 % to 55.1 % (−28.7 points, 95 % CI [−29.3, −28.1]) and residual exposure quadruples. That is a lower bound on what avoidance could achieve at 90 m, not an estimate of it: these routes were planned against 60 m cones and only re-scored, a router given wider cones would find different clean routes, and a full re-route at each radius—most of a day apiece—has not been done. But if real fields of view are half again as long as assumed, more than a third of the routes counted as clean here are not. The sweep re-scores the map's wedges; the registry's discs share the radius and have no span to sweep, and the directed scoring of Section 5.2 is their sensitivity check.

8.5 A single avoidance strength

All results use the strongest setting, 0.01: a watched road costs a hundred times its usual weight, close to but short of prohibition, so "zero cameras" means the optimiser found a clean path, not that one was guaranteed to exist. Intermediate settings are the practically interesting ones for a shipped application, and in informal tests whether one took effect depended on local camera density; a single slider-to-multiplier mapping is unlikely to serve.

8.6 Sample scope

All fifty states and the District of Columbia, but within-state commutes only (`od_main`), so cross-border metropolitan areas—New York/New Jersey, Philadelphia, Washington, Kansas City, St Louis, Portland—are systematically truncated at the state line. Several are exactly the dense multi-jurisdiction areas where the question is most pointed. The District is the extreme case: most people who work there live in Maryland or Virginia, so its sample describes only commutes with both ends inside the District. Alaska's flows are from 2016 and Michigan's from 2021, the latest LODES publishes; the other 49 jurisdictions' are from 2022.

LODES pairs a home with a job's establishment, and one pair in eight in the sample lies more than 100 km apart (Section 4.6). Those pairs make the tail, and they raise the price of refusal more than the share exposed: within 100 km, 76.0 % [75.6–76.5] of commuters pass a reader rather than 78.6 %, but avoiding costs a median 1.70 min [1.66–1.77] rather than 2.29 min, and the mean commute passes 4.90 cameras rather than 6.24. The registry's effect is much the same at that range: on the map alone the figures are 72.5 % and 1.20 min.

8.7 Past maps record mapping, not deployment

The trend of Section 6.4 is a trend in what had been mapped. The edit history gives the day each camera entered the map to the second; it does not give the day the camera went up, which OpenStreetMap records for 47 of the 142,257. A camera first mapped in 2026 may have stood since 2021, and the 128-fold growth of the map cannot be read as a rate of deployment. Absolute figures date accordingly. The relative claims are the ones to carry forward, and they held on every map: the cheapness of refusal—under a minute per camera evaded on all twelve routed maps, though rising on each since July 2024—and the near-proportionality of measured exposure to the map across all thirty-five. What the maps cannot establish is that the current one is complete, and the merged map of the present has no past: the registry's record dates are not installation dates (Section 4.2).

8.8 Free-flow travel times

Routing uses free-flow speeds with no traffic, time-of-day, or turn-restriction penalties, so both travel times are optimistic. The *difference* between them—the quantity actually reported—is more robust than either absolute, since both share the bias.

9 Ethics and dual use

The study uses only public data. The camera locations are volunteers' observations from public places, already published in OpenStreetMap under an open licence, and the vendor's device records, already published by an independent researcher [30], which we use in aggregate and do not redistribute; the commute flows and neighbourhood attributes are Census Bureau aggregates, published with the Bureau's confidentiality protections and attached to census tracts rather than to people; the portal counts are published by the police departments themselves. No individual is identified, no one's travel was observed, and the paper adds no camera location to the public record.

The routing engine is dual-use, and we weighed that before publishing it. It lets anyone decline to be photographed on the way to work, including someone with reason to evade a lawful investigation. Three considerations decided the matter. The capability is not new: the camera map is public, and several services already route drivers around it [1, 5, 15]. The result it rests on bounds the investigative value of fixed cameras against a motivated subject whether or not the tool exists (Section 7); what such a network reliably records is the unmotivated majority. And the measurement itself—how much surveillance an ordinary trip passes, what declining it costs, and how fast both are growing—is information a public debate about the technology needs and has not had. The app is released under GPLv3 and the server and study code under the MIT licence, and the study's routing server is not exposed to the internet (Section 3).

10 Reproducibility

Code, configuration, and the derived outputs behind every table and figure—per-commute results on every routed date, per-state tables, and every analysis report quoted in the text—are in the project repository, `https://github.com/KaraZajac/TRAMES`, under `research/`. Fixed seed `20260725` throughout, for both sampling and bootstrap. All external data sources are public and require no API key. Beyond the routing server, the analysis needs Python with NumPy, Shapely and Matplotlib; pyosmium for the road sample and the history dump; and pyogrio, which bundles GDAL, only to read the HPMS geodatabase. The repository's README names the script behind each step, from the camera fetch and the edit-history extraction through sampling, routing, the past-date runs and the analyses, to these figures and tables.

The 19.2 GB OSM extract, the four 14.7 GB graph caches, the 163 GB OSM full-history dump, the 1.3 GB of TIGER/Line boundaries and the 5.6 GB HPMS geodatabase are not redistributed; the acquisition and import scripts rebuild all of them. The transparency-portal counts behind the coverage check come from a live service and are kept as retrieved, in `research/data/validation/`. The registry snapshot is not redistributed: it is available from its publisher [30], and the scripts that read it, merge it with the map and build its discs are in the repository. Commuter weights come from `research/out/sampling_frame.json` and `sample_draws.csv`, written by the sampler.

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